Makoto Yamada

Introduction to Machine Learning 2026

2026/09/04

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Course Information

- This course consists of lectures and hands-on.
- For hands-on, we will use Python with Google Colab.

- Mid-term exam: 30% (written exam)
- Final project: 70%

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Schedule (Tentative)

Mon, Sep 7	**Introduction to Machine Learning**	Slides (PDF) Hands-on
Wed, Sep 9	**Linear Algebra for ML**	Slides (PDF) Hands-on
Mon, Sep 14	**Probability and Maximum Likelihood Estimation**	Slides (PDF) Hands-on
Wed, Sep 16	**Linear Regression**	Slides (PDF) Hands-on
Mon, Sep 21	***National Holiday***	*Respect for the Aged Day — no class*
Wed, Sep 23	***National Holiday***	*Autumnal Equinox Day — no class*
Mon, Sep 28	**Classification**	Slides (PDF) Hands-on
Wed, Sep 30	**Nonlinear Regression**	Slides (PDF) Hands-on
Mon, Oct 5	**Midterm Exam**	
Wed, Oct 7	**Review of Midterm Exam**	
Mon, Oct 12	***National Holiday***	*Sports Day — no class*
Wed, Oct 14	**Feature Selection and Sparsity**	Slides (PDF) Hands-on
Mon, Oct 19	**Dimensionality Reduction** PCA, CCA, t-SNE	Slides (PDF) Hands-on
Wed, Oct 21	**Introduction to Deep Learning**	Slides (PDF) Hands-on
Mon, Oct 26	***No Class***	*No class this week*
Wed, Oct 28	***No Class***	*No class this week*
Mon, Nov 2	**Introduction to Graph Neural Networks**	Slides (PDF) Hands-on
Wed, Nov 4	**Introduction to Self-Supervised Learning**	Slides (PDF)
Mon, Nov 9	**Introduction to Optimal Transport**	Slides (PDF)
Wed, Nov 11	**Project 1**	
Mon, Nov 16	**Project 2**	
Wed, Nov 18	**Project 3**	
Mon, Nov 23	***National Holiday***	*Labor Thanksgiving Day — no class*
Wed, Nov 25	**Project 4**	
Mon, Nov 30	**Final Presentation 1**	
Wed, Dec 2	**Final Presentation 2**	

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Mid-term Exam & Final project

- Mid-term Exam (Written exam, 30% of grading)
  - Basic mathematics: linear algebra, probability
  - Programming: Basic manipulation, numpy etc.
- Final project (70% of grading)
  - Work on an original machine learning related project.
  - Try to get non-public data at OIST (hopefully, from your potential supervisor)
  - You can use ChatGPT and all the public datasets. However, if there is no originality, the score can be lower. (e.g., use someone’s Github project)
  - Will be evaluated in report and presentation.

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Question

What is artificial intelligence?

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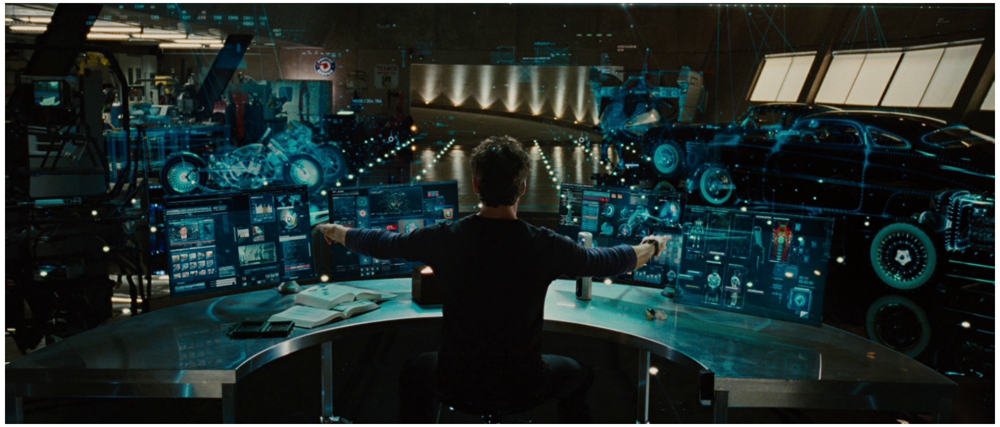
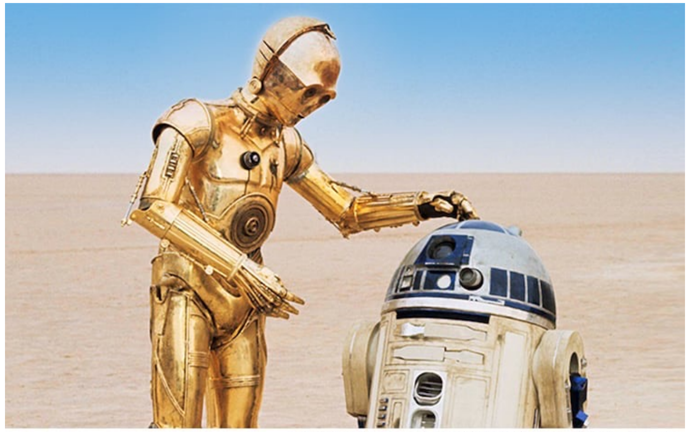
Artificial General Intelligence, AGI





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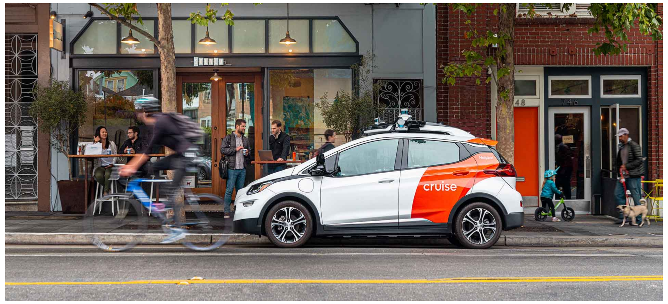
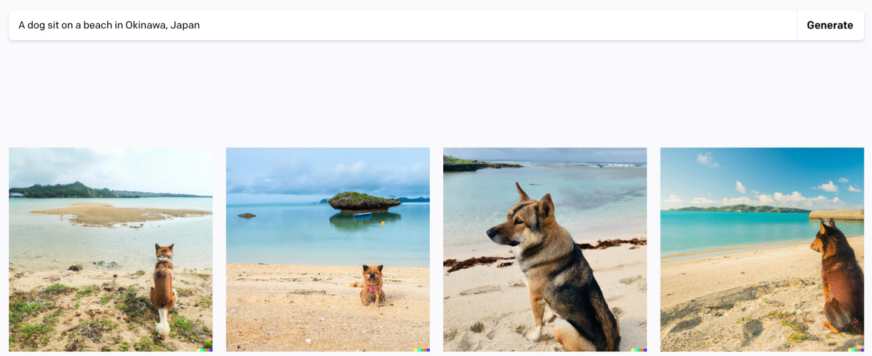
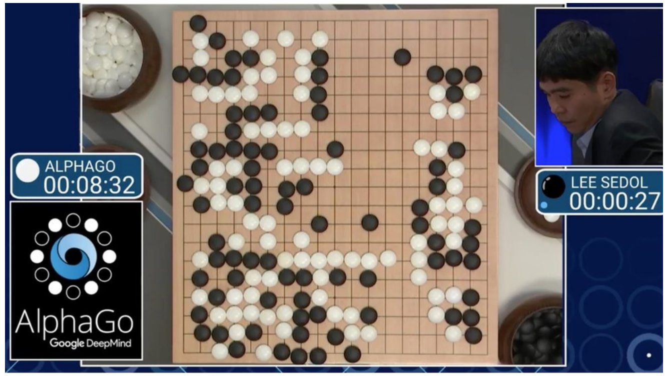
Artificial Intelligence

- General-purpose artificial intelligence (artificial intelligence that can think like a human)
- Specialized Artificial Intelligence (AI specializing in individual fields or domains)
  - Autonomous driving
  - Machine translation
  - Protein folding prediction
  - Image generation
  - Etc.







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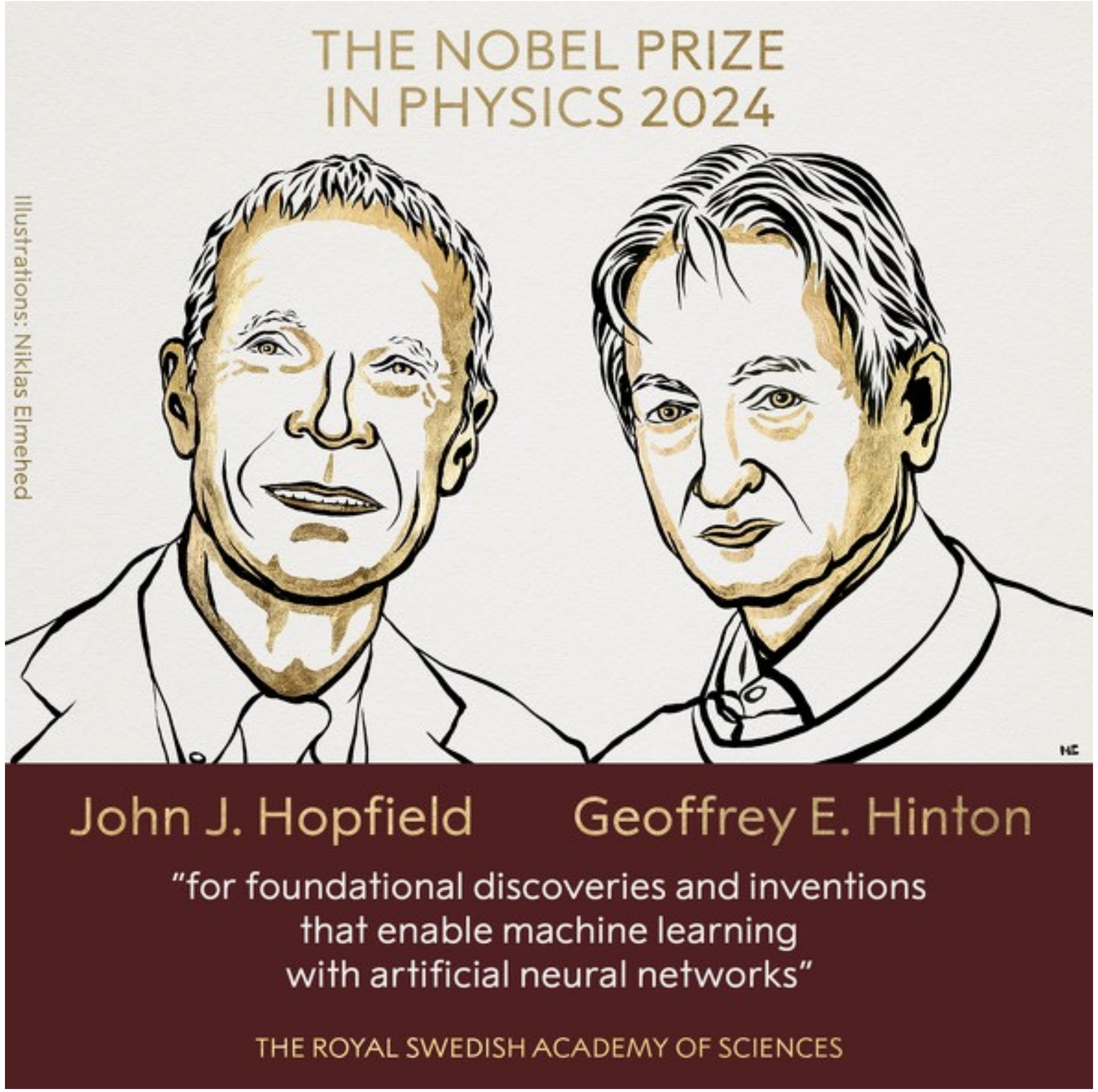
Nobel Prize in Physics (2025)



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Nobel Prize in Chemistry (2025)

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Recent trends in AI (Last week)

- Large Language Models (LLM) & Multimodal Agents

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Ten advances in mathematics and theoretical computer science (by Astra)

- **High-dimensional sphere packing.** New upper bounds on sphere-packing density down to the Cohn–Elkies threshold.
- **Binary and spherical codes.** Exponentially improved bounds on the maximum size of binary codes at any prescribed minimum distance, with analogous results for high-dimensional spherical codes.
- **Non-sofic groups.** A construction establishing the existence of non-sofic groups, addressing a central open question in group theory.
- **Connes's rigidity conjecture.** Disproof of a longstanding conjecture that certain groups are uniquely determined by their von Neumann algebras.
- **Arithmetic circuit complexity.** New lower bounds for computing the permanent using arithmetic circuits and formulas, including an arithmetic-formula lower bound of order $n^4/\log n$.
- **Quantum parallel repetition.** An exponential parallel repetition theorem for general two-player quantum games, extending a foundational principle from classical complexity theory.
- **Closest vector problem.** Polynomial-factor hardness of approximation for the closest vector problem, a foundational lattice question related to post-quantum cryptography.
- **Ehrhart's volume conjecture.** Determining, in every dimension, the maximum possible volume of a convex body whose centroid is its only interior lattice point.
- **Multicolor Ramsey numbers.** A superexponential lower bound for multicolor triangle Ramsey numbers, resolving Erdős problem 183.
- **Extremal number conjectures.** Results on the compactness and degeneracy conjectures in extremal graph theory, resolving Erdős problems 146 and 180.

The total number of tokens needed to find solutions to these problems would cost roughly $2,000 at Sol API rates.

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Recent trends in AI

- World Models

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Embodied AI

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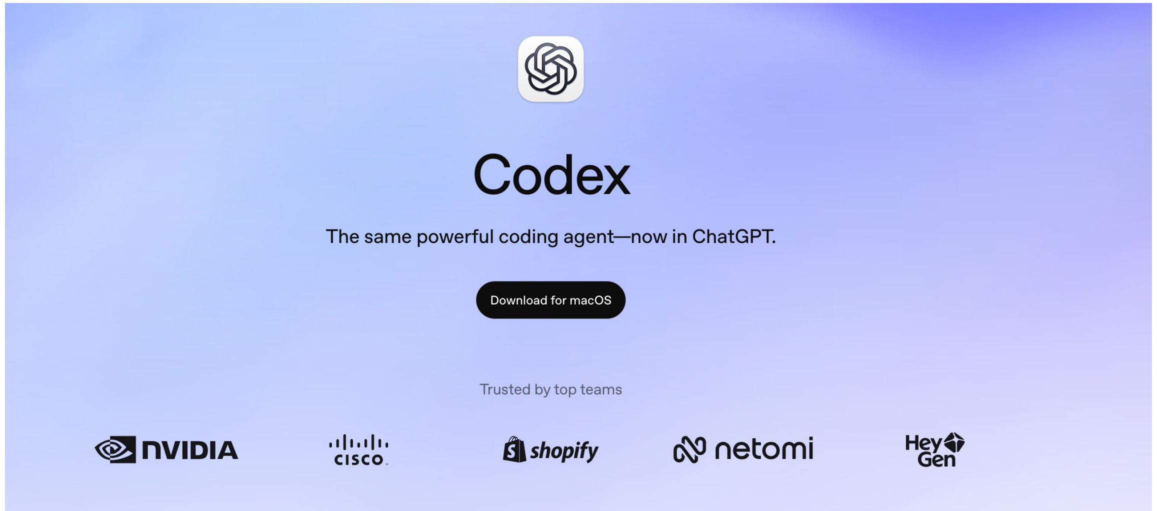
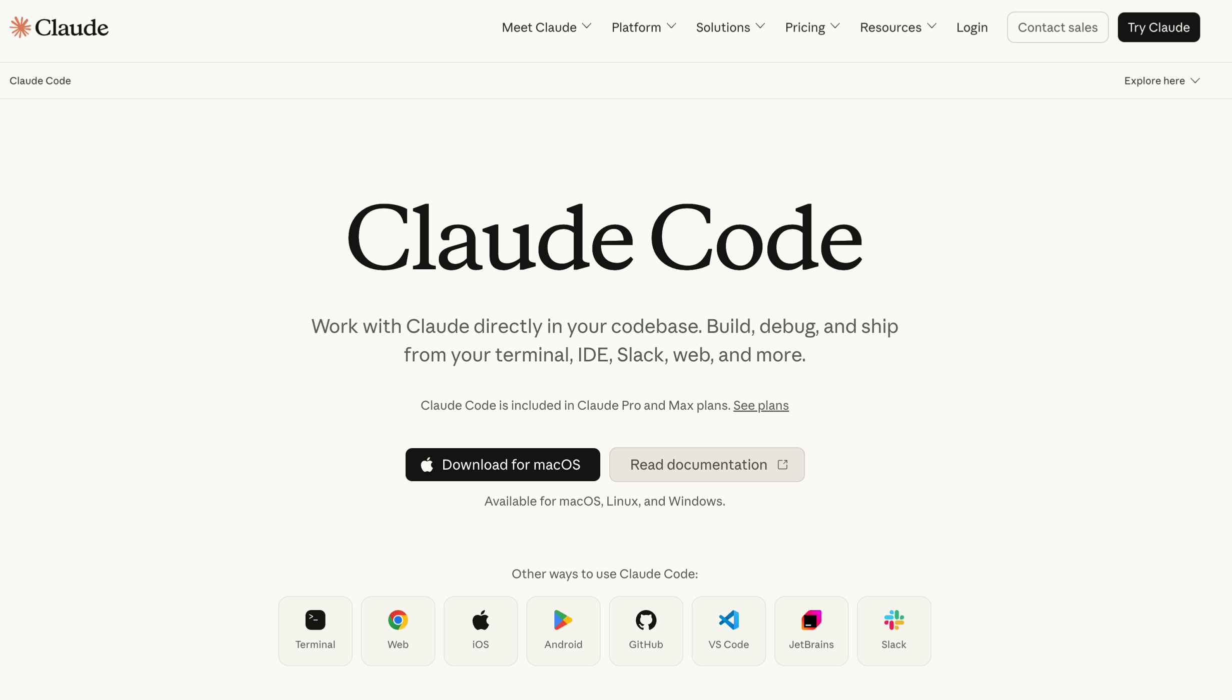
AI agents (coding)

- Codex, Claude code, etc.





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What we should learn in this course?

- The foundations of these methods are based on developments over the past 60 years.
- So, this is a mixture of history of AI (ML) and learn some basics of AI.
- Note: The content of this course (in particular for the recent applications) will become obsolete quite rapidly ☹
- But, the foundation of ML will not change ☺

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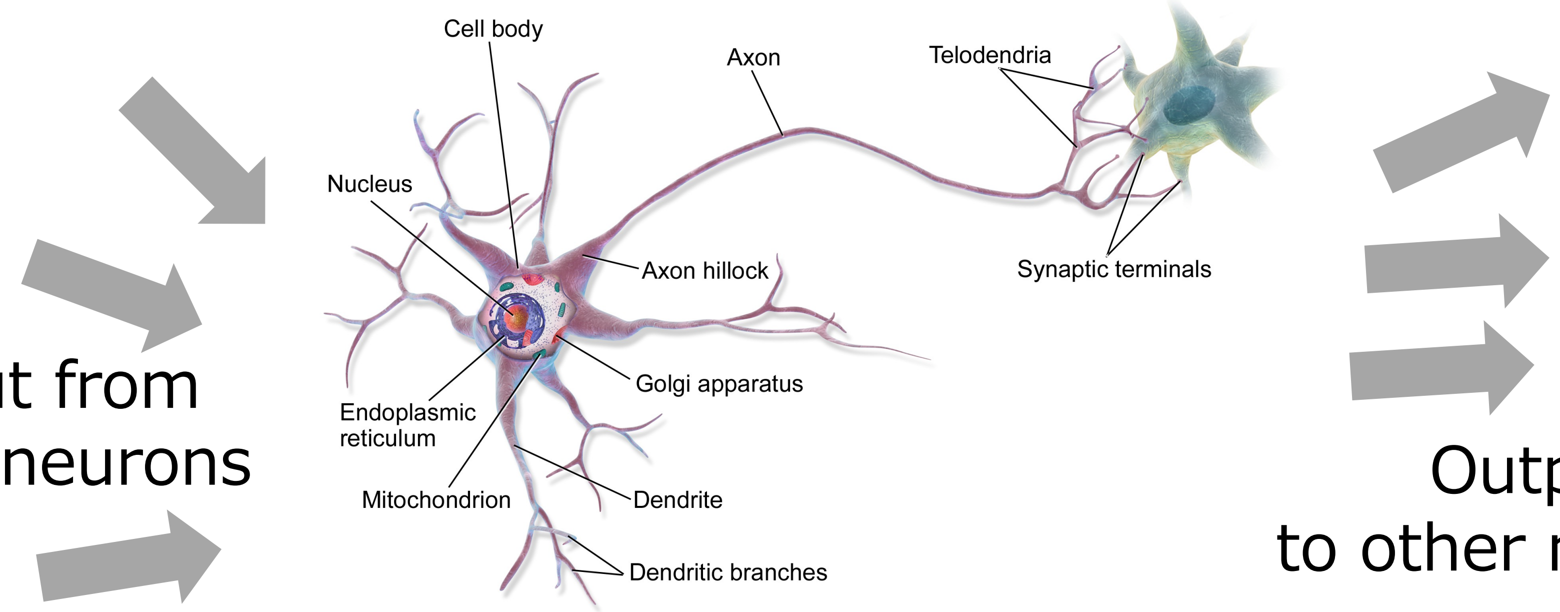
Neurons: the 'parts' responsible for information processing in the brain

- The brain is made up of many neuronal connections
- Information from other neurons is sorted, integrated and sent out.

Input from
other neurons



Output
to other neurons

https://en.wikipedia.org/wiki/Neuron

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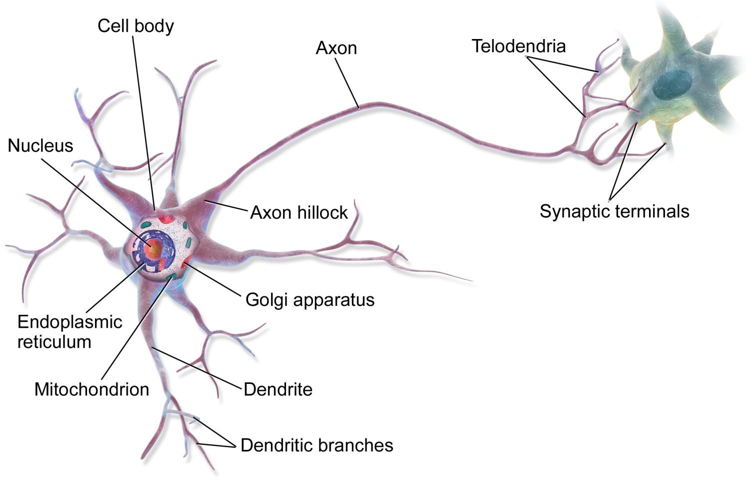
Classifier and neuron similarity

- Both classifiers and neurons are components that weigh and aggregate each element of the input to compute an output.
- Both can be seen as simple computational units
- Our brain is a collection of many neurons



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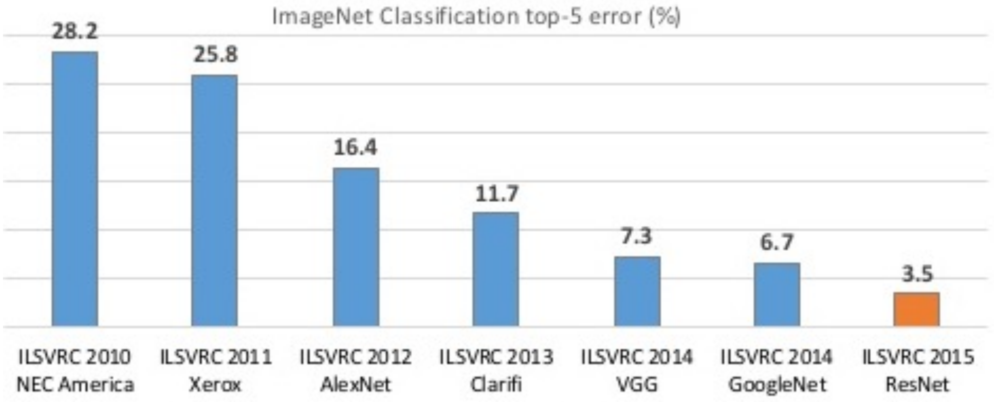
Deep learning era

- 2012 Deep learning wins a competition for image identification performance, beating the previous record by more than 10 percentage points → Gains prominence.
- Companies such as Google and Meta (Facebook) invest heavily in deep learning
- Deep learning also takes center stage in research and development trends



Microsoft presentation slide

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History of Large Language Models (LLM)

- Transformer (2017)
- GPT (Generative Pre-trained Transformer) (2018)
  - Model size (0.117B)
- GPT-2 (2019)
  - Model size (1.5B)
- GPT-3 (2020)
  - Model size (175B)
- ChatGPT (GPT-3.5, GPT-4)
  - Human feedback
- Gemini, Claude, Mistral, etc.
- Reasoning models: OpenAI o1 / o3, GPT-5
- Gemini 2.x / 3, Claude 4 / 5 (Sonnet, Opus, Haiku)
- Open-weight models: Llama, DeepSeek-R1, Qwen
- DeepSeek-R1 (early 2025): strong reasoning performance at much lower training cost → intensified efficiency race

Many of ML researchers working on this topic…

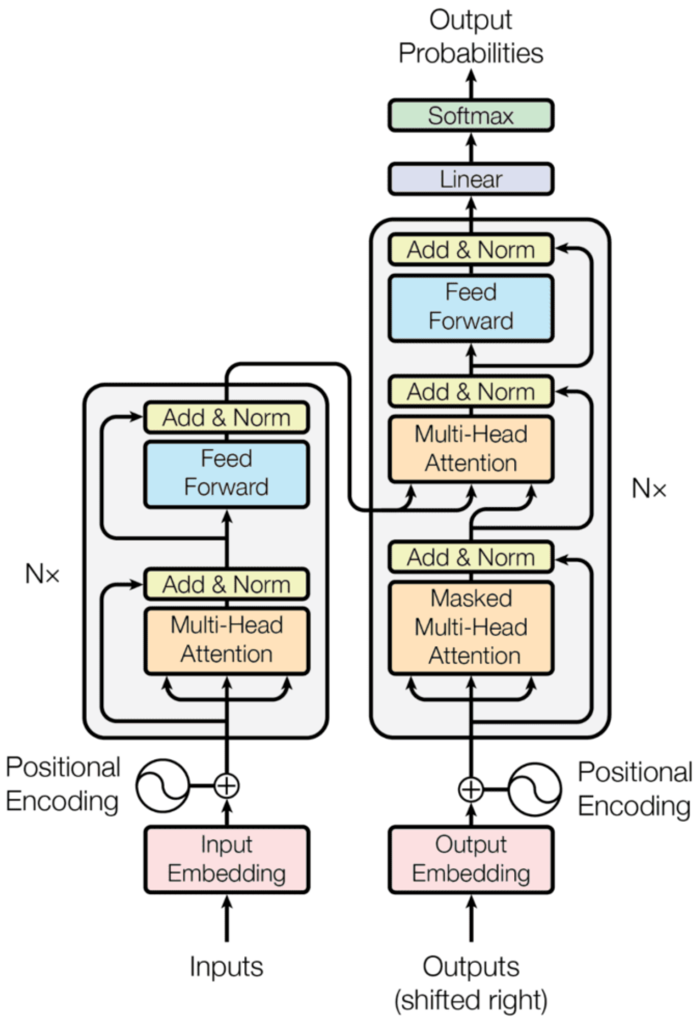

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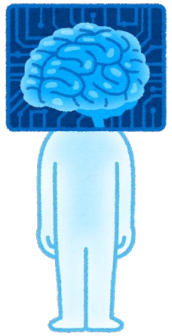
Image classification (Object classification)

Input



Output

Cat

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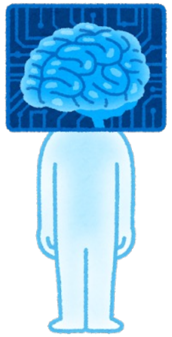
Image classification (Face identification)

Input



Output

Makoto

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Object Detection

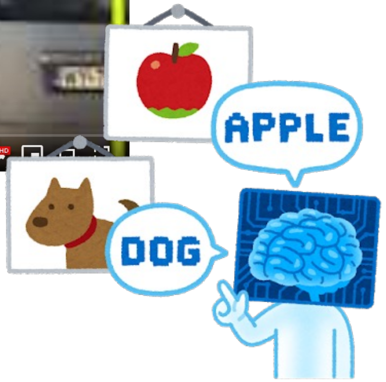

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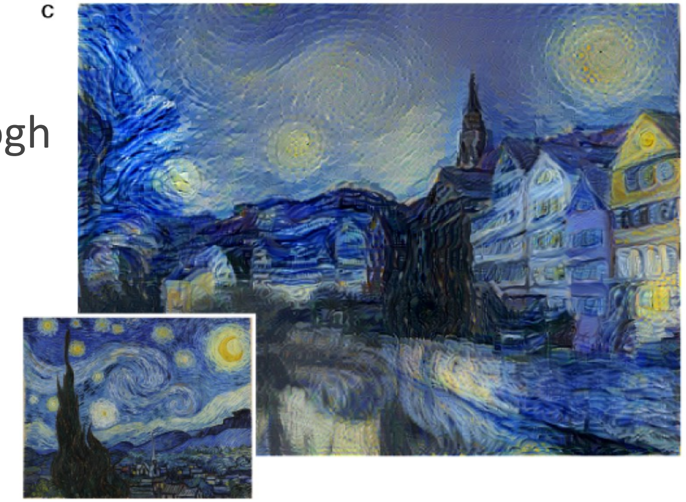
Image transformation

Converted to Van Gogh style

C



Converted to Munch style

D

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Deep Fake (https://en.wikipedia.org/wiki/Deepfake)

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Image generation

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Image generation (DALL-E)

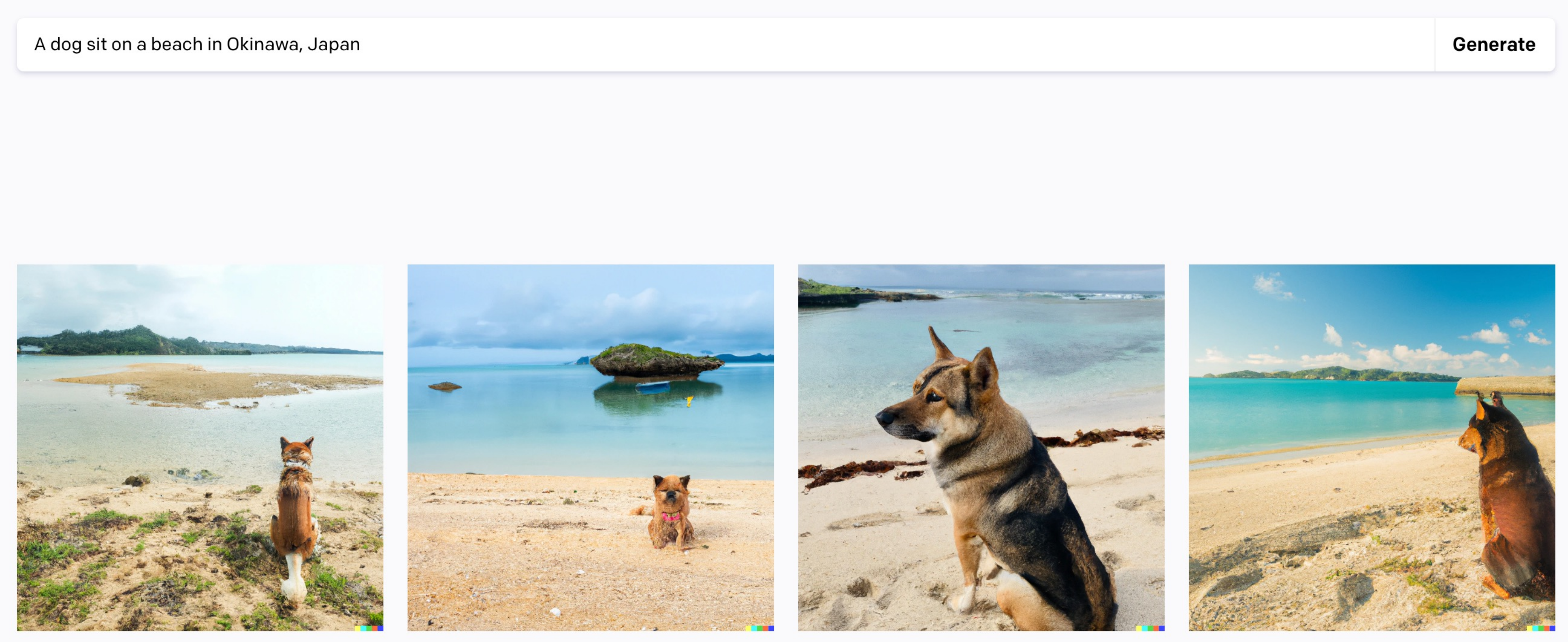

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Image generation (Midjourney)

- https://www.sciencefocus.com/future-technology/midjourney/

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Video Generation (SORA)

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Veo3 (2025)

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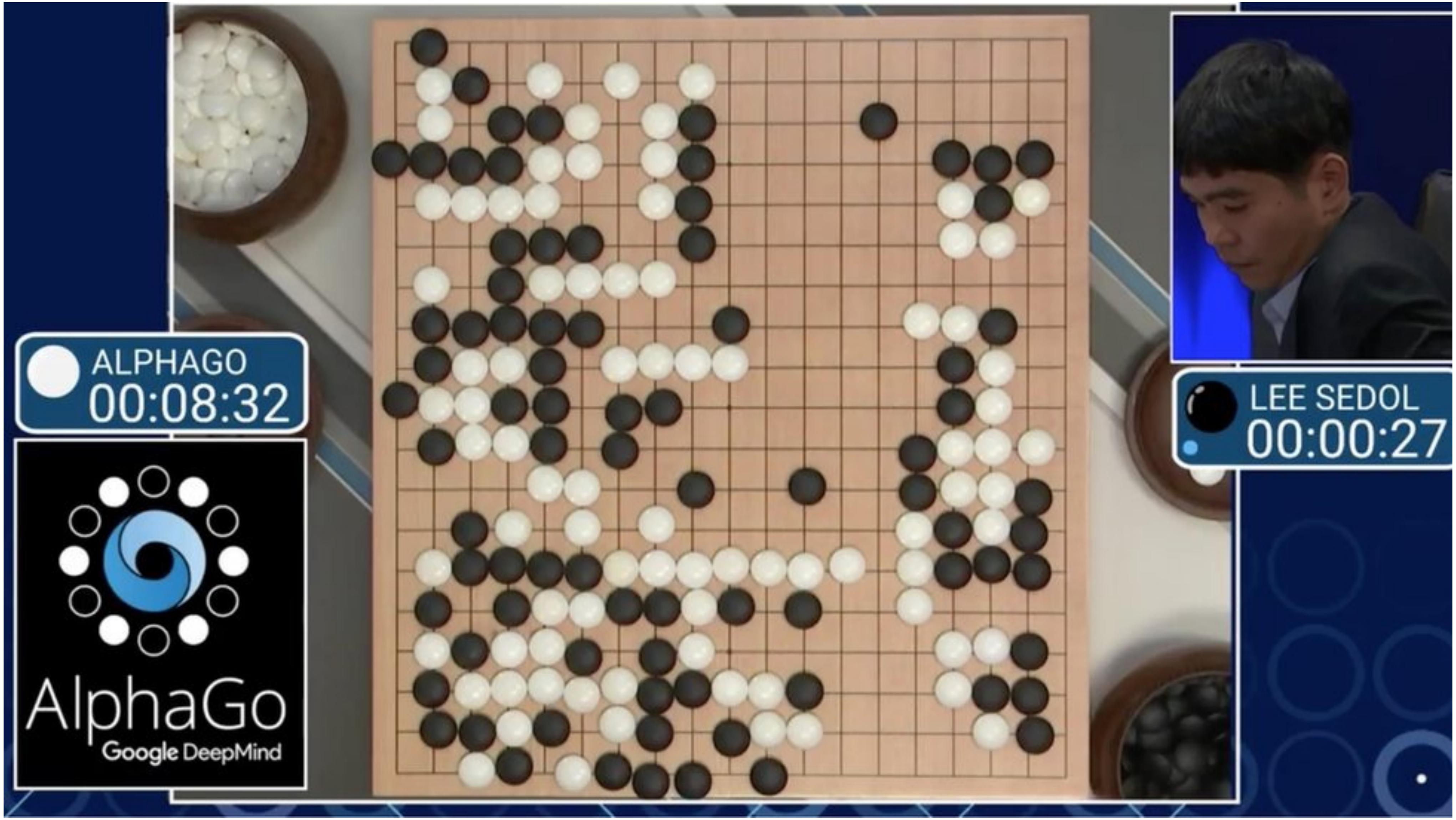
Alpha-Go



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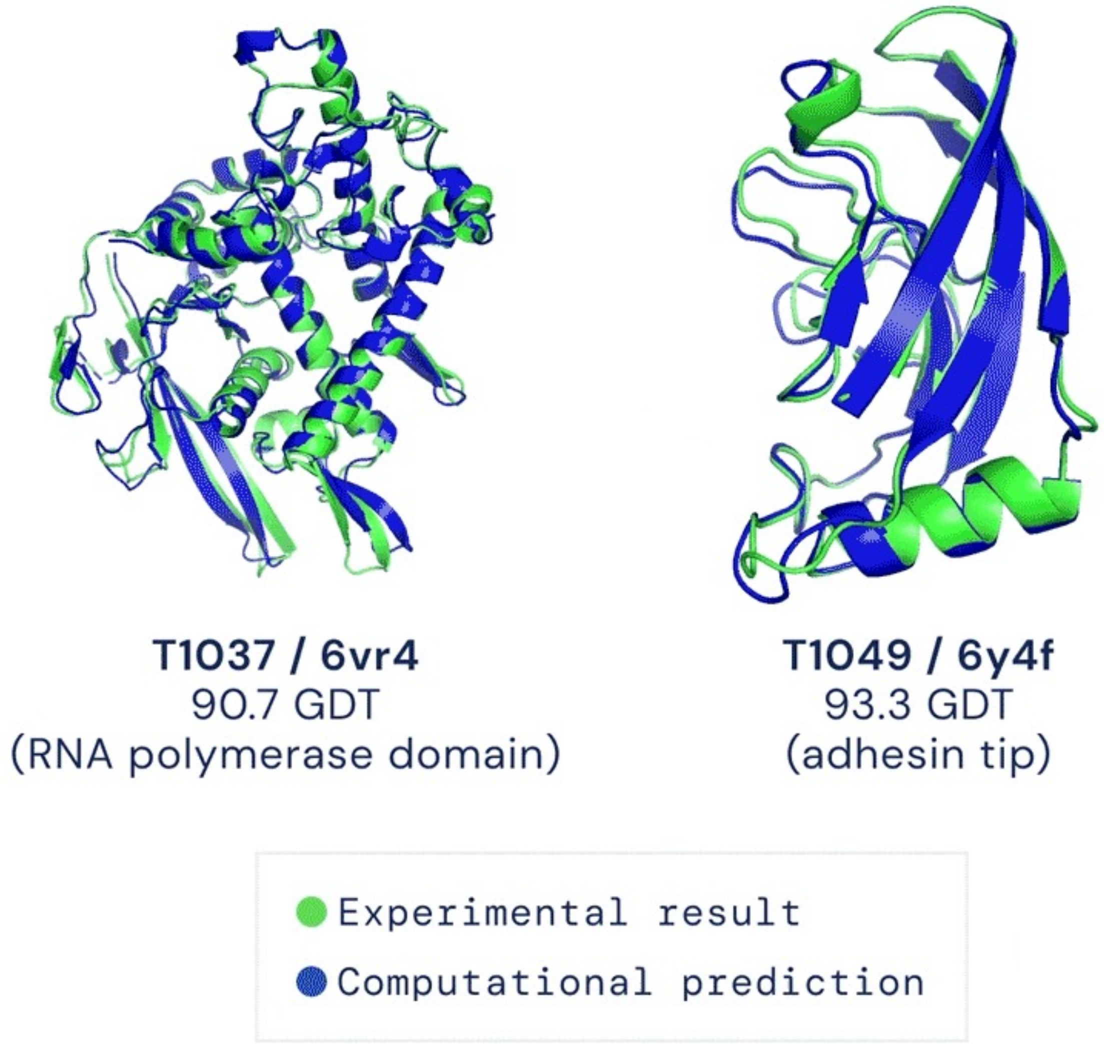
Alpha-Fold



https://www.deepmind.com/blog/alphafold-a-solution-to-a-50-year-old-grand-challenge-in-biology

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Alpha-Tensor

$$\begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \\ 5 & -1 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & -1 & 0 \\ 5 & 1 & -1 \\ -2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -2 & -1 & 0 \\ 11 & -2 & -1 \\ 1 & -6 & 1 \end{bmatrix}$$

$$3 \times 2 + 1 \times 5 + 0 \times -2 = 11$$

Example of the process of multiplying two 3x3 matrices.

https://www.deepmind.com/blog/discovering-novel-algorithms-with-alphatensor

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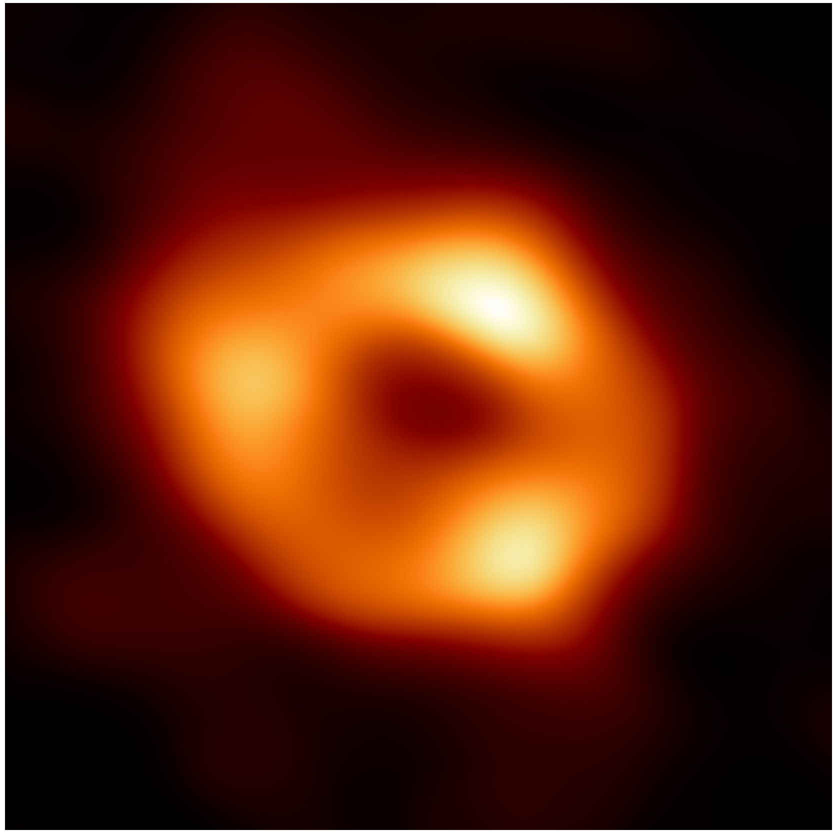
Visualizing Black hole



https://eventhorizontelescope.org/blog/astronomers-reveal-first-image-black-hole-heart-our-galaxy

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Machine Translation

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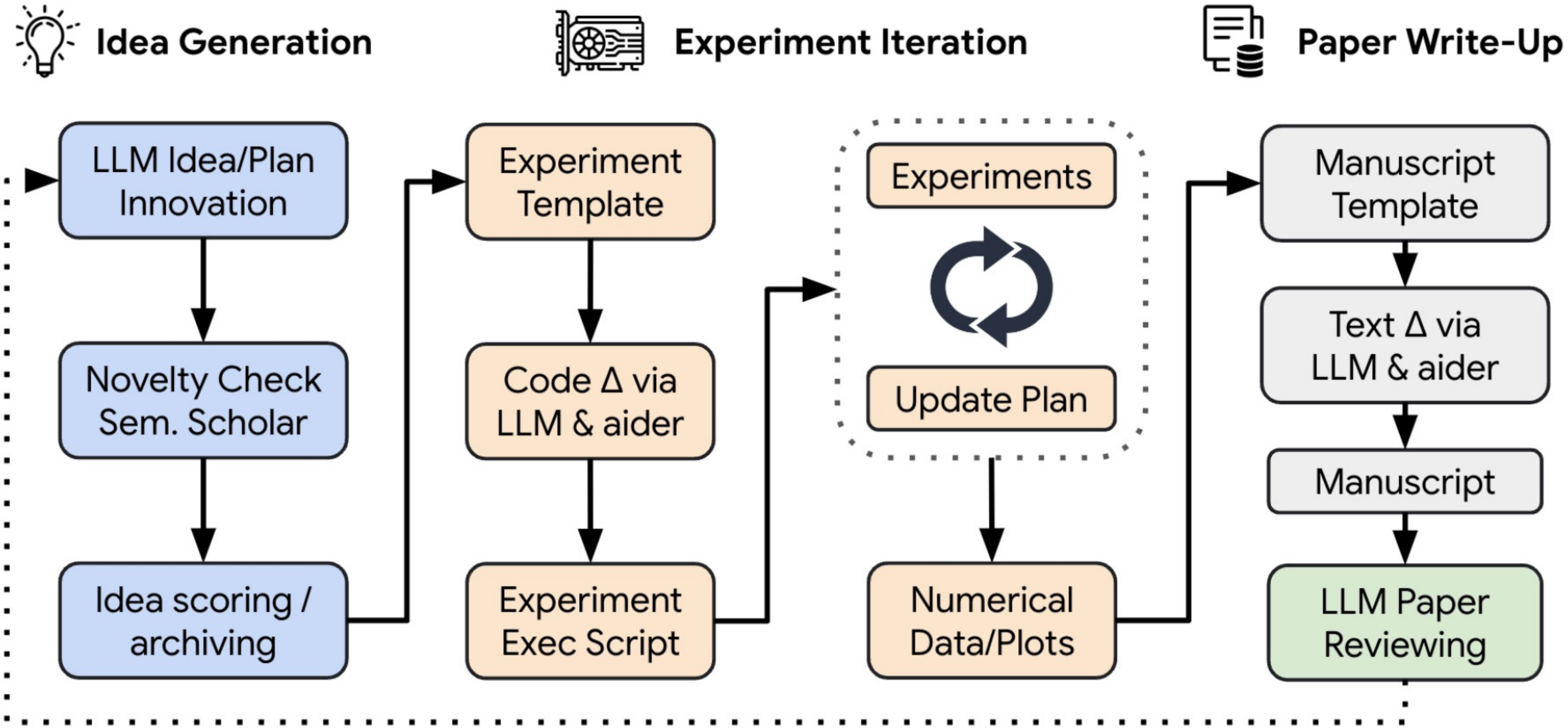
AI Scientist



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AI Scientist

1 of 11 Automatic Zoom

DualScale Diffusion: Adaptive Feature Balancing for Low-Dimensional Generative Models

Anonymous authors
Paper under double-blind review

Abstract

This paper introduces an adaptive dual-scale denoising approach for low-dimensional diffusion models, addressing the challenge of balancing global structure and local detail in generated samples. While diffusion models have shown remarkable success in high-dimensional spaces, their application to low-dimensional data remains crucial for understanding fundamental model behaviors and addressing real-world applications with inherently low-dimensional data. However, in these spaces, traditional models often struggle to simultaneously capture both macro-level patterns and fine-grained features, leading to suboptimal sample quality. We propose a novel architecture incorporating two parallel branches: a global

DualScale Diffusion: Adaptive Feature Balancing for Low-Dimensional Generative Models

[Link to Full PDF] [Link to Code]

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Machine Assisted Proof

Machine assisted proof

Terence Tao *

February 10, 2024

Mathematicians have relied on upon computers (human, mechanical, or electronic) and machines to assist them in their research for centuries (or even millennia, if one considers early calculating tools such as the abacus). For instance, ever since the early logarithm tables of Napier and others, mathematicians have known the value of constructing large data sets of mathematical objects to perform computations and to make conjectures. Legendre and Gauss used extensive tables of prime numbers compiled by human computers to conjecture what is now known as the prime number theorem; a century and a half later, Birch and Swinnerton-Dyer similarly used early electronic computers to generate enough data on elliptic curves over finite fields to propose their own celebrated conjecture on these objects. And many readers have no doubt taken advantage of one of the broadest mathematical data sets of all, the Online Encyclopedia of Integer Sequences, which has generated numerous conjectures and unexpected connections between different areas of mathematics, as well as serving as a valuable mathematical search engine

computers has been in *scientific computation*, which is heavily used nowadays to numerically solve differential equations and dynamical systems, or to compute the statistics of large matrices or linear operators. An early example of such computation arose in the 1920s, when Hendrik Lorentz assembled a team of human computers to model the fluid flow around the *Afsluitdijk* - a major dam then under construction in the Netherlands; among other things, this calculation was notable for pioneering the now-standard device of floating point arithmetic. But modern computer algebra systems (e.g., Magma, SAGEMath, Mathematica, Maple, etc.), as well as more general-purpose programming languages, can go well beyond traditional "number-crunching"; they are now routinely used to perform symbolic computations in algebra, analysis, geometry, number theory, and many other branches of mathematics. Some forms of scientific computations are famously unreliable due to round-off errors and instabilities, but one can often replace these methods with more rigorous substitutes (for instance, replacing floating point arithmetic with interval arithmetic), possibly at the expense of increased runtime or memory usage.

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ChatGPT

MY

Teach artificial intelligence so simply that even a baby can understand it.

Alright, little one! Imagine you have a super-smart friend called AI, which stands for Artificial Intelligence. Just like you learn from your experiences, AI learns from lots of information it reads and sees. Imagine it's like AI's special food that makes it smarter.

You know how when you play with your toys, you figure out which ones make you happy or sad? AI does something similar. It looks at pictures, reads stories, and listens to people to understand things. Then, it tries to guess what might happen next based on what it knows.

Think about your favorite game where you guess what's hiding under the cups. AI does that too! It looks at patterns in the pictures and words it sees and tries to guess what they mean. Just like you learn to recognize your family and friends' faces, AI learns to recognize things too.

Imagine you have a magic drawing book that turns your scribbles into real things. AI kind of has a magic drawing book too, but instead, it creates pictures, music, and even stories. It looks at all the things it learned and tries to make something new and exciting, just like when you create your art.

So, Artificial Intelligence is like a clever friend who learns from information, guesses what might happen, recognizes things like faces, and creates new stuff. Just like you, AI keeps getting smarter as it learns more!

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ChatGPT

- Basic idea: Predicting next words

Example 1: The prefecture of Okinawa is _______

Example 2: A famous aquarium in Okinawa is _______

Very easy for humans, but super difficult to do on a computer...

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ChatGPT

Example 1: The prefecture of Okinawa is Naha city.

- P(Naha| The prefecture of Okinawa is) = 0.85
- P(Okinawa| The prefecture of Okinawa is) = 0.10
- P(Onna| The prefecture of Okinawa is) = 0.05

Example 2: A famous aquarium in Okinawa is Churaumi aquarium.

- P(DMM-Kariyushi| A famous aquarium in Okinawa is) = 0.3
- P(Churaumi| A famous aquarium in Okinawa is) = 0.7

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ChatGPT

- How to achieve?
  - Predicting the probability of the next word from the word sequence obtained so far.
- Probability (Degree of certainty that an event will occur, often expressed as P(event).)
  - For six-sided dice, event is defined as 1, 2, 3, 4, 5, 6
  - If the dice is not skewed, P(dice = 1) = 1/6
  - Sum to 1. P(dice=1) + P(dice=2) + … + P(dice=6) = 1
- Joint probability (Probability of multiple events occurring)
  - Probability of two dice being rolled.
  - P(dice1=1,dice2=5)

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ChatGPT

- Conditional probability (Probability of another certain event 2 occurring under the condition that event 1 occurs.)
  - P(Event2 | Event1)
- Bayes theorem
  - P(Event2 | Event1) = P(Event1, Event2)/P(Event1)

Example: Consider the probability of travelling in Okinawa Prefecture.

Where to go (Event): Churaumi, OIST

- P(Churaumi) = 0.8, P(OIST) = 0.2
- P(Churaumi,OIST) = 0.4, P(Churaumi, Churaumi) = 0.1
- P(OIST| Churaumi) = P(Churaumi, OIST)/P(Churaumi) = 0.4/0.8 = 0.5

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ChatGPT

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Quiz

- What is the estimated cost of one ChatGPT training?
  1. 100 $
  2. 10000$
  3. 1M $
  4. More than 1M$

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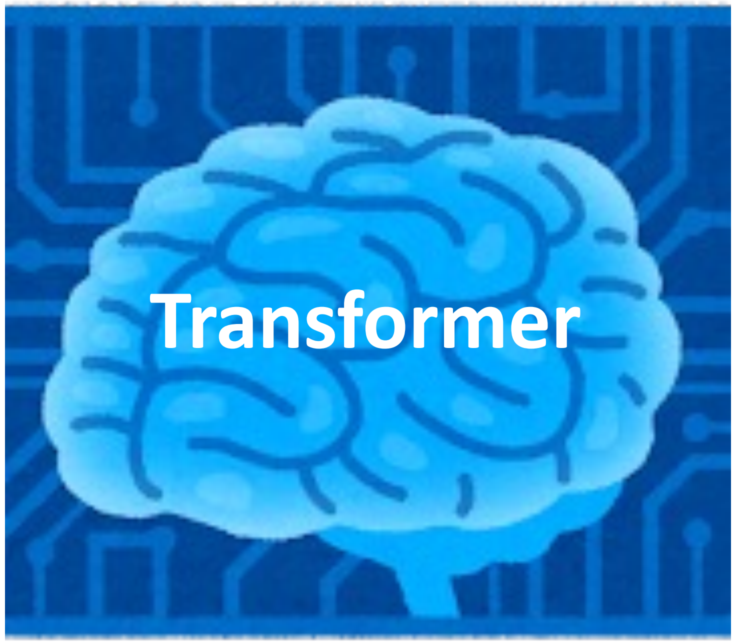
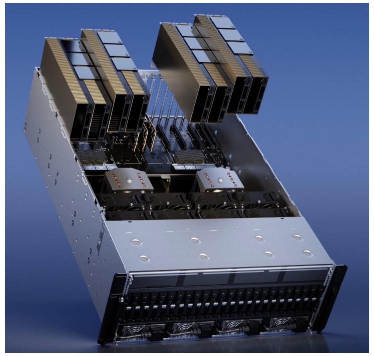
ChatGPT

Large documents
(570GB in 2020; multi-TB+ today)





Graphic Processing Unit
(GPU, A100x10000 in 2020;
far larger clusters today)

Conditional probability P(Y | X)

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History of ChatGPT

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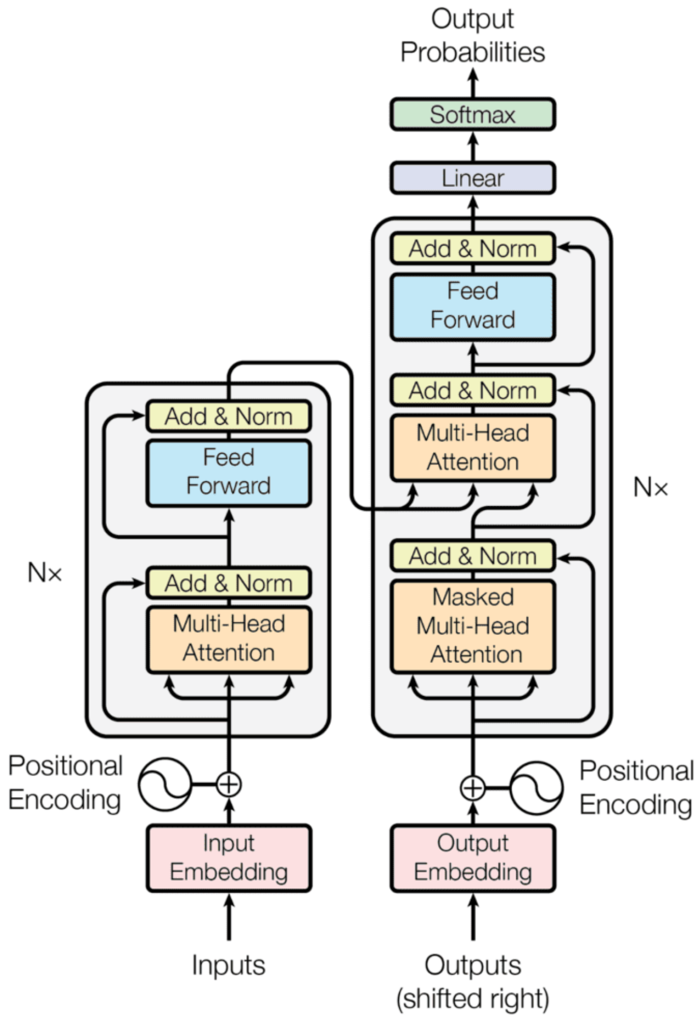

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Prompting

- Considering the input for LLM

MY

Answer three numbers of the following sequence: 0, 1, 1, 2, 3, 5, ...

The sequence you've provided is the Fibonacci sequence. To continue the sequence, the next three numbers would be:

8, 13, 21.

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Prompting

MY

Please write a title and abstract of story based on Okinawa in 30 words

Title: "Whispers of Shisa: Unearthing Okinawa's Mysteries"

Abstract: In the enchanting landscapes of Okinawa, a curious archaeologist delves into ancient ruins, awakening forgotten legends and encountering the protective spirits of Shisa, setting off an adventure of heritage, courage, and rebirth.

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Prompting

MY

You are a famous writer. Please write a title and abstract of story based on Okinawa in 30 words

Title: "Azure Bonds: Chronicles of Okinawa"

Abstract: Against Okinawa's backdrop, disparate lives converge—a marine biologist, an elderly storyteller, and a war veteran—forging connections that illuminate forgotten histories, ecological wonders, and the human spirit's resilience.

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Hallucination

MY

Please list three artificial intelligence researchers in the Okinawa Institute of Science and Technology

As of my last update in September 2021, here are three researchers associated with the Okinawa Institute of Science and Technology (OIST) who were involved in artificial intelligence research:

1. **Kenji Doya**: Kenji Doya is a neuroscientist and researcher at OIST who has worked on computational neuroscience and machine learning. His research often focuses on understanding the neural mechanisms of decision-making and learning, and he has applied AI techniques to study brain functions. Please note that his research focus may extend beyond AI alone.
2. **Bernd Kuhn**: Bernd Kuhn is a researcher at OIST who has worked on machine learning and data analysis. His research interests include developing statistical and machine learning methods for analyzing complex data, including data from neuroscientific experiments. Like other researchers, his work might not be solely centered on AI but could encompass related fields.
3. **Yohei Yamada**: Yohei Yamada is a researcher at OIST who has been involved in projects related to robotics and AI. His work has included developing algorithms for robot learning and control. His research often falls within the intersection of robotics and machine learning.

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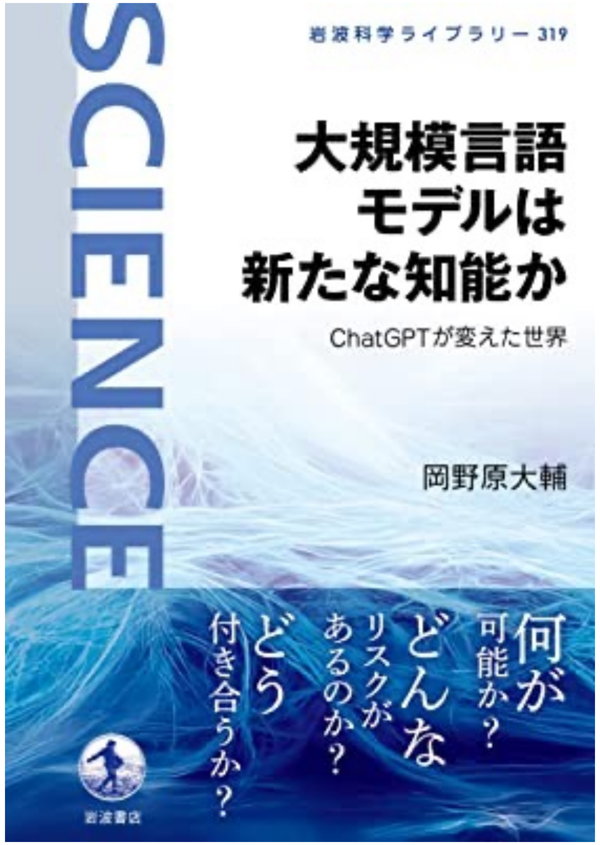
Risk of ChatGPT

- Credibility of information
- Private dialogue information
- Values and prejudices
- Difficulty in proving identity
- Changes in the way work is done
- Concentration of power among a few large-model
- Environmental and energy costs of AI
- Impact on jobs and the labor market



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What we do not know yet

- Why transformer works?
- Relationship between transformer and brain
- Efficient model update
- Efficient model (Brain: ~20W; large-scale AI training/inference: orders of magnitude more energy)
- A way to remove hallucination
- …

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To understand ChatGPT

- Statistics (Probability)
- Linear algebra (Transformer model)
- Derivative (Transformer model training)
- Optimization
- …

Mathematics (and programming) is important to understand the latest artificial intelligence techniques!

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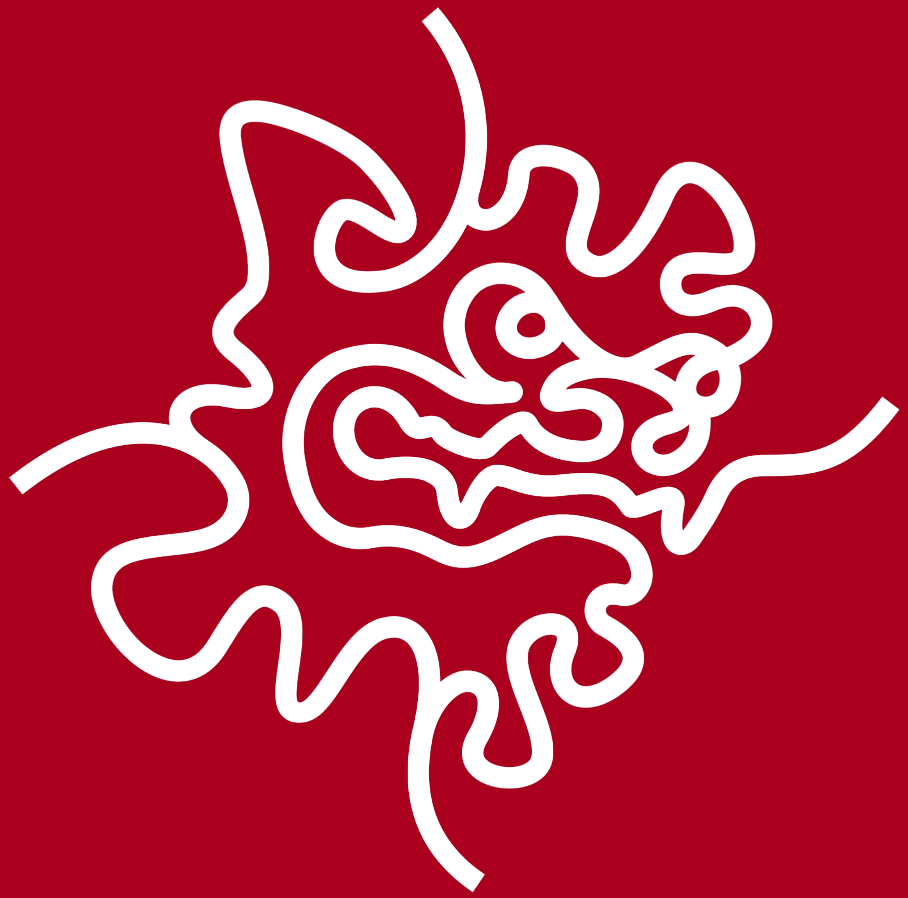

Thank you!